

ADANI CEMENT · INDUSTRIAL AI ENGAGEMENT

Golden Operating Window for Rotary Kiln

A joint engagement to identify and establish the optimum operating window for the rotary kiln, enabling consistent clinker production and improved operational performance, delivered by Faclon Labs on AWS.

Objective

Golden Operating Window for Rotary Kiln — Golden Batch, Optimum Control Limits of Operation

Develop a plant specific of Golden Batch of Control limits of operation from historical operating conditions reflecting the most efficient performance at different productivity levels in terms SHC, TSR, Clinker Quality. These Golden Batch of Control limits form the navigational benchmark of the operating personnel to replicate and improve upon for a more efficient operation.

Clinker Production

Specific Heat Consumption (SHC)

Thermal Substitution Rate (TSR)

Clinker Quality

Deliverables

Golden Operating Window

Golden Batch — Optimum Control Limits

- Plant-specific Golden Operating batch for all agreed key kiln process and quality parameters.
- Historical benchmark of best-performing operating periods and the operating control limit ranges to be validated jointly with plant process experts and Faclon SME.

Live Benchmarking Dashboard

- Real-time comparison of live operation against the Golden Operating Window. (Golden Batch Control limits at different productivity levels)
- Parameter-wise deviation monitoring and overall Golden Window compliance. (overall compliance of operation to Golden batch control limits)

Root Cause Analysis & Recommendations

- Root Cause Analysis for selected parameters in case while deviating from golden operating control limits such as SHC, TSR, C3S, FcaO, SAT, TAT, BZT and other agreed KPIs.
- Prioritised, operator-friendly recommendations to restore optimize operation towards the Golden Batch Operating Window Control limits.

Scope of Work

Historical Data Assessment

- Collect and validate a minimum of 12 months of historical process and quality data.
- Review data availability across the kiln, preheater, calciner, cooler, fuel systems and laboratory quality data.

Golden Operating Window Development

- Clean and prepare historical operating data by removing shutdowns, start-ups and abnormal operating periods.
- Identify historically best-performing operating periods based on productivity, SHC, TSR and clinker quality.
- Derive the recommended operating control limit ranges for the key kiln process and quality parameters that consistently delivered the best performance.

Online Benchmarking & Operator Guidance

- Integrate real-time process data and quality data (where available) into the Golden Operating Window framework.
- Continuously compare live operating conditions against the Golden Operating Window.
- Detect and highlight deviations from the recommended operating ranges.
- Perform Root Cause Analysis (RCA) for selected parameters such as SHC, SAT and BZT.
- Generate operator-friendly recommendations to guide the process back towards the Golden Operating Window.

NOTES

- A minimum of 12 months of historical process and quality data will be available for model development.
- Real-time process and quality data required for online benchmarking will be made available by the plant at the required frequency.
- RCA and recommendations will initially be developed for selected critical KPIs and can be expanded in future phases.
- Recommendations are advisory in nature and intended to support operator decision-making. The identified operating windows control limits, root cause analysis and recommendations shall be jointly reviewed and validated with the plant operations team, Adani process experts and Faclon SMEs. Any operational changes should be implemented by the plant team after appropriate technical verification to achieve the desired process improvements.**